

Credit Card Default Auto-ML Report — lgbm (sub)

 2026-05-19 23:21:13 UTC

모형 lgbm

roc_auc = 0.7746

BEST(catboost) = 0.7796

Δ vs best = -0.0050

1 점수 요약 (본 모형 vs best)

metric	lgbm (현재)		catboost (BEST)		Δ(Test)
	Train	Test	Train	Test	
roc_auc	0.8273	0.7746	0.8099	0.7796	-0.0050
ks	0.5022	0.4248	0.4660	0.4362	-0.0114
lift	3.2530	3.1123	3.4037	3.1274	-0.0151
accuracy	0.8188	0.8137	0.8308	0.8177	-0.0040
precision	0.7399	0.7061	0.7133	0.6589	+0.0472
recall	0.2791	0.2698	0.3931	0.3640	-0.0942
f1	0.4054	0.3904	0.5069	0.4689	-0.0785
pr_auc	0.5999	0.5430	0.6194	0.5588	-0.0158

2 Feature Importance — lgbm

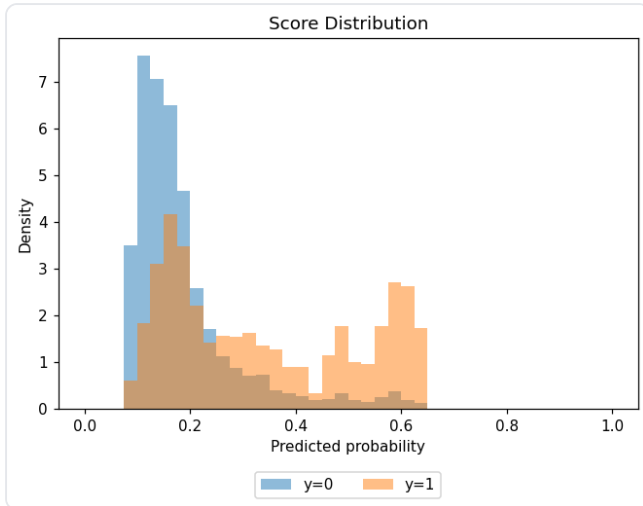
relative · top 19 · 전체 분포는 feature_importance.csv

Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.4705	16	PAY_5	0.0154
2	PAY_2	0.1389	17	BILL_AMT6	0.0149
3	LIMIT_BAL	0.0435	18	PAY_6	0.0123
4	BILL_AMT1	0.0390	19	SEX	0.0027
5	PAY_AMT3	0.0353			
6	PAY_3	0.0349			
7	PAY_AMT2	0.0284			
8	PAY_AMT1	0.0256			
9	BILL_AMT2	0.0243			
10	PAY_AMT4	0.0211			
11	PAY_4	0.0209			
12	AGE	0.0196			
13	PAY_AMT6	0.0184			
14	PAY_AMT5	0.0180			
15	BILL_AMT5	0.0163			

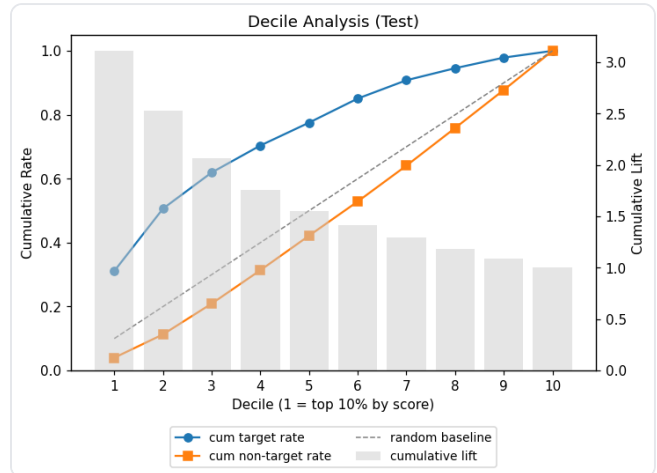
3 모델 상세 — lgbm

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.477 ~ 0.630	600	413	187	413	187	600	10.0%
2	0.297 ~ 0.477	600	259	341	672	528	1200	20.0%
3	0.222 ~ 0.297	600	150	450	822	978	1800	30.0%
4	0.188 ~ 0.222	600	112	488	934	1466	2400	40.0%
5	0.169 ~ 0.188	600	95	505	1029	1971	3000	50.0%
6	0.152 ~ 0.169	600	100	500	1129	2471	3600	60.0%
7	0.137 ~ 0.152	600	76	524	1205	2995	4200	70.0%
8	0.120 ~ 0.136	600	50	550	1255	3545	4800	80.0%
9	0.104 ~ 0.120	600	44	556	1299	4101	5400	90.0%
10	0.086 ~ 0.104	600	28	572	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4524	149
True 1	969	358

4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature_selection.csv)

Feature		Selection Frequency	채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외

5 학습 설정 + 튜닝 — lgbm

키	값
tuned	YES n_trials = 15 , best_value = 0.7849
best_iteration (avg over folds)	154
params (최종)	{'objective': 'binary', 'metric': 'auc', 'verbose': -1, 'n_estimators': 800, 'learning_rate': 0.021471858797027, 'num_leaves': 85, 'min_data_in_leaf': 55, 'feature_fraction': 0.7779717392187493, 'bagging_fraction': 0.6052561262871233, 'reg_alpha': 0.07115953064092197, 'reg_lambda': 7.860792250906604, 'min_gain_to_split': 0.1471710061864095}